

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2- Concept Mapping (5 QUESTIONS)

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	30% of CIA	Total Marks:	50 (5 Questions × 10Marks)
CO1 Statement:	Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.		
Student Name:	P.V.D.Nagendra Reddy	Reg. No.:	192472399.
Section / Batch:	SLT D / 2024	Date:	14/07/26

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Code of Conduct

I Nagendra (192472399) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nagendra Reddy.

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision, Features of Computer Vision, and Levels of Computer Vision	Computer Vision pipeline, features, and processing levels	1	10
Digital Image Fundamentals	Image representation, resolution, and image quality	2	10
Image Sensing and Acquisition, Sampling and Quantization	Digital image acquisition, sampling, and quantization process	3	10
Image Formation Models	Real-world scene to digital image formation	4	10
Applications of Image Processing and Computer Vision	Computer Vision capabilities and real-world applications	5	10
GRAND TOTAL:			50 Marks

Annexure A –Concept mapping

1. Construct a comprehensive and well-organized concept map illustrating the complete pipeline from real-world scene capture to digital image representation. Identify all major concepts, establish meaningful relationships among intermediate transformations and dependencies, integrate insights from at least five recent literature sources, and present the map with clear hierarchy, linking phrases, and logical organization.

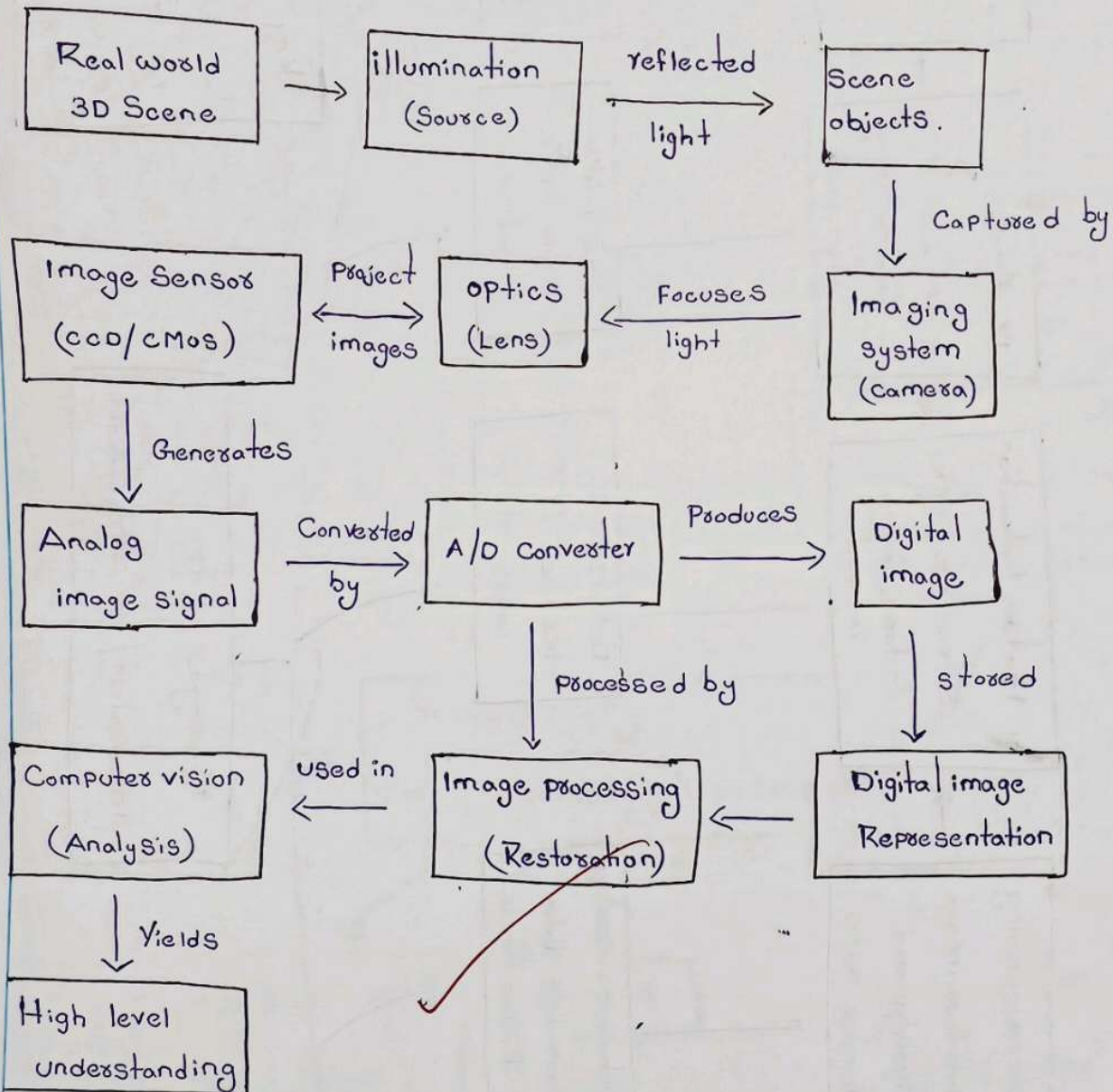
2. Develop a hierarchically structured concept map illustrating the major stages of visual data handling in a Computer Vision system. Clearly show the relationships among acquisition, preprocessing, feature extraction, analysis, and interpretation stages, explain their contributions to meaningful outputs, incorporate findings from at least five literature sources, and ensure clarity and readability.

3. Create a detailed concept map explaining the relationships among image resolution, spatial resolution, intensity resolution, and image quality. Clearly illustrate cause-and-effect relationships, discuss their impact on image usability in different applications, integrate evidence from at least five research articles, and organize the concepts logically.

4. Design an analytically organized concept map illustrating the factors influencing image acquisition conditions, including illumination, sensor characteristics, motion, environment, and optics. Show how these factors affect downstream image processing and application performance, integrate relevant literature findings, and present the map in a visually coherent manner.

5. Construct an original concept map illustrating the relationship between Computer Vision system capabilities and the selection of appropriate vision-based applications. Identify major system capabilities, establish their influence on application effectiveness, integrate perspectives from multiple literature sources, and present the concepts using a clear hierarchical structure.

① Computer Vision Pipeline : From scene To digital image

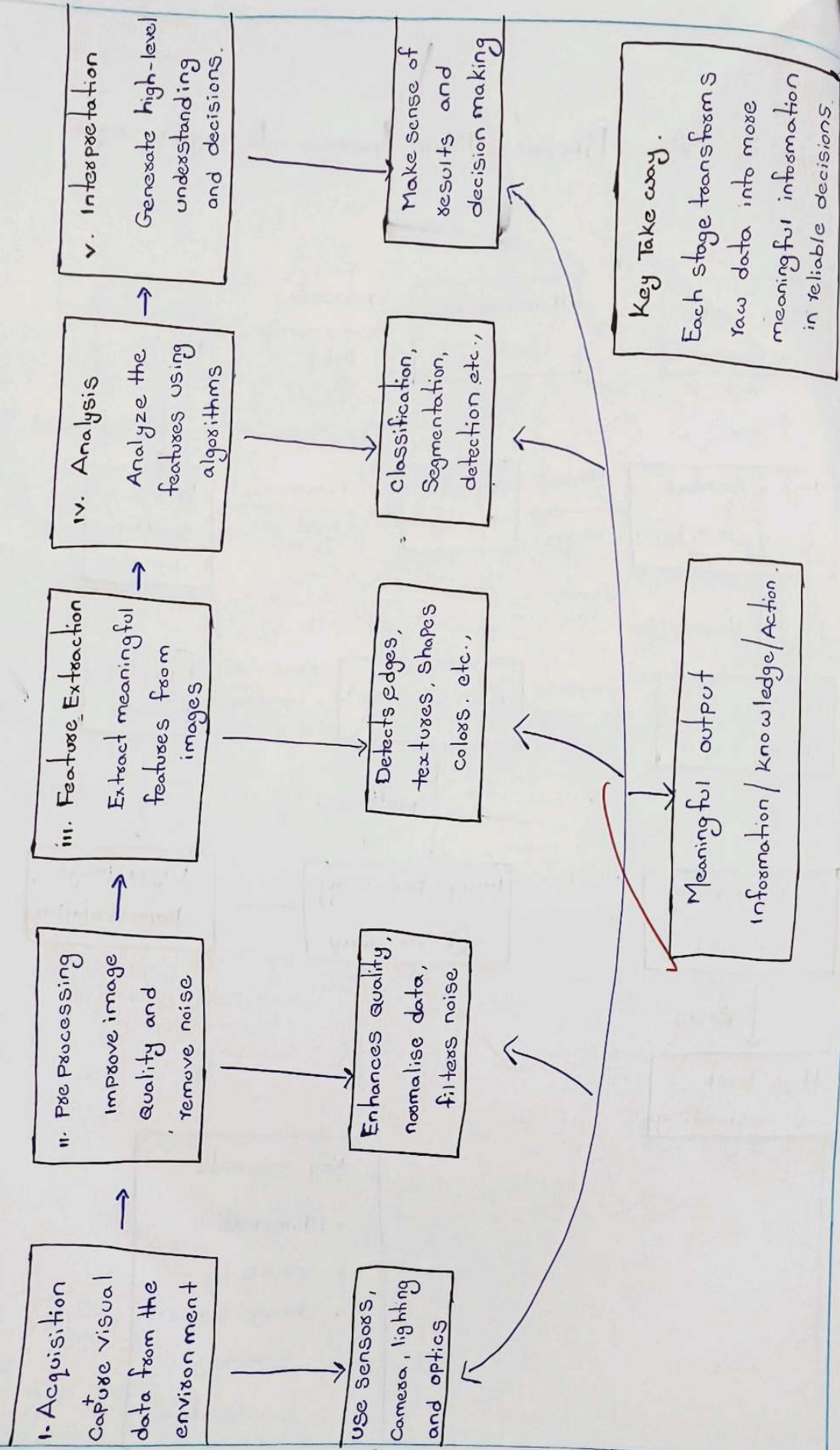


Key concepts

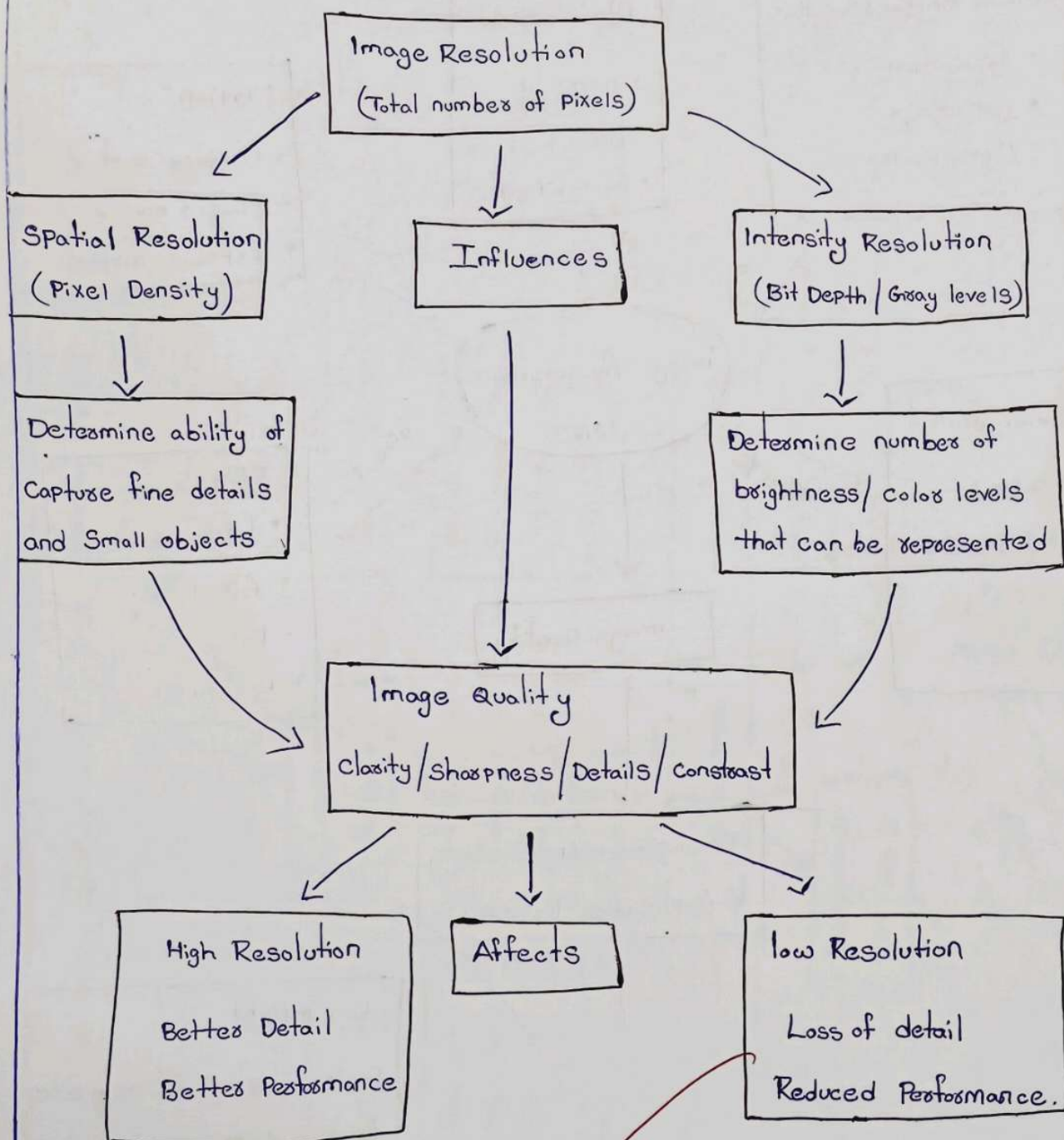
- illumination
- optics
- Image Sensor
- Sampling
- Quantization.

②

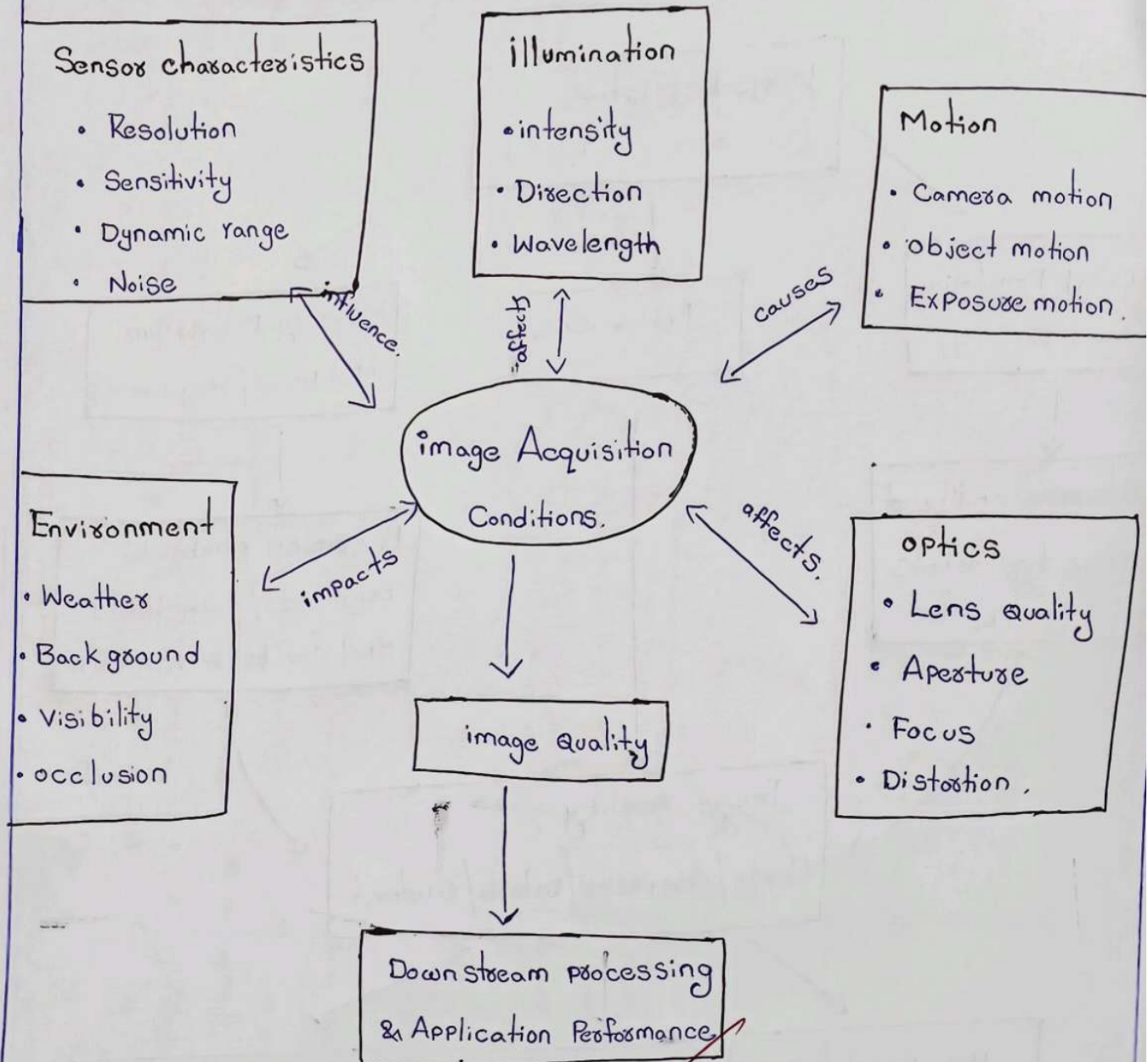
Major stages of visual data handling in Computer vision.



Relationship Among image Resolution, Spatial Resolution, Intensity Resolution & Image Quality.



④ Factors Influencing image Acquisition Conditions



Key Insight

Proper control of acquisition factors improves image quality and boosts overall system performance.

⑤ Computer Vision Capabilities and Selection of Vision - Based Applications .

